

Testing the limits of logical reasoning in neural and hybrid models

Compositionality and Reasoning in AI and Cognitive Science
Workshop

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Our contributions

- ▶ We present a framework, inspired by compositional tests (Hupkes et al., 2019), for systematically evaluating generalization in logical reasoning over natural language fragments (Pratt-Hartmann, 2004).

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Our contributions

- ▶ We present a framework, inspired by compositional tests (Hupkes et al., 2019), for systematically evaluating generalization in logical reasoning over natural language fragments (Pratt-Hartmann, 2004).
- ▶ Using syllogistic logic, one of the smallest logical fragment, we evaluate neural assistance in symbolic proof construction and identify significant limitations in generalization.
- ▶ We present a neuro-symbolic syllogistic prover that uses neural guidance for proof construction, achieving efficient symbolic search and robust, interpretable reasoning despite limited neural generalization.

Syllogistic logic

Well-formed formulas		
<i>Aab</i>	$a \subseteq b$	All <i>a</i> are <i>b</i>
<i>Eab</i>	$a \cap b = \emptyset$	No <i>a</i> are <i>b</i>
<i>Iab</i>	$a \cap b \neq \emptyset$	Some <i>a</i> are <i>b</i>
<i>Oab</i>	$a \not\subseteq b$	Some <i>a</i> are not <i>b</i>

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- ▶ An *A-chain* ($Aa - b$) represents either a formula Aab or the sequence $Aac_1, Ac_1c_2, \dots, Ac_{n-1}c_n, Ac_nb$ (for $n \geq 1$)
- ▶ The **negation** of a formula F is denoted as $\overline{F}$

$$\begin{array}{l|l} \overline{Aab} = Oab & \overline{Iab} = Eab \\ \overline{Oab} = Aab & \overline{Eab} = Iab \end{array}$$

Knowledge Base

Definition

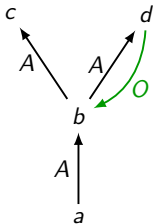
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Example: graph representation.



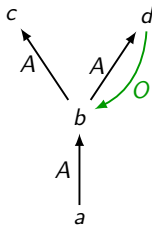
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$$\mathcal{KB} = \{Aab, Abc, Abd, Odb\}$$

Remark: we generate $\mathcal{KB}$ s that are **consistent** (no contradictions) and **non-redundant**, in the sense that each hypothesis admits only one minimal **proof**.

Derivation rules (Smiley, 1973)

Definition

A *derivation* ∇ is one of the following three types:

- (i) Every $F \in \mathcal{KB}$ is a derivation from $\mathcal{KB}$

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$$\frac{\nabla' \quad \nabla''}{\frac{Aab \quad Abc}{Aac}} \text{ (r1)}$$

$$\frac{\nabla' \quad \nabla''}{\frac{Aab \quad Ebc}{Eac}} \text{ (r2)}$$

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- (iii) *Proof by contradiction*: where ∇' is a derivation from $\mathcal{KB} \cup \{\overline{H}\}$ and ∇'' is a derivation from $\mathcal{KB}$.

$$\frac{\nabla' \quad \nabla''}{\frac{F \quad \overline{F}}{H}} \text{ (iii)}$$

Types of syllogisms

Definition (Syllogism)

Let $\mathcal{F}$ be a set of formulas (*premises*) and let F be a formula (*conclusion*). We write $\mathcal{F} \vdash F$, to denote that F is provable (or derivable) from $\mathcal{F}$, i.e., there exists a proof of F from $\mathcal{F}$.

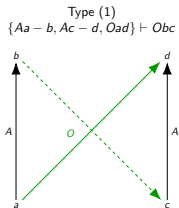
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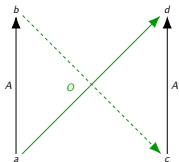
Type	Syllogism
(1)	$\{Aa - b, Ac - d, Oad\} \vdash Obc$
(2)	$\{Aa - b\} \vdash Aab$
(3)	$\{Aa - b, Ac - d, Aa - e, Ede\} \vdash Obc$
(4)	$\{Aa - b, Aa - c\} \vdash Ibc$
(5)	$\{Aa - b, Ac - d, Ae - f, Iae, Edf\} \vdash Obc$
(6)	$\{Aa - b, Ac - d, Ebd\} \vdash Eac$
(7)	$\{Aa - b, Ac - d, Iac\} \vdash Ibd$

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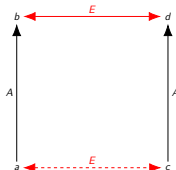


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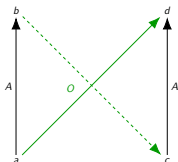


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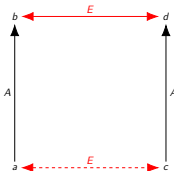


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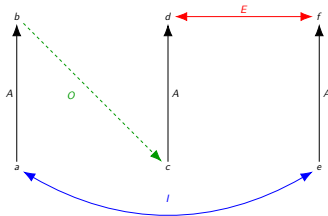
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Type (5)
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A semantic approach: from antilogisms to syllogisms

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Theorem 1 (Smiley, 1973)

Every antilogism is of the following form

$\{Aa - b, Oab\},$

$\{Aa - b, Aa - c, Ebc\},$ or

$\{Aa - b, Ac - d, Iac \text{ (or } Ica), Ebd\}.$

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Theorem 2 (Smiley, 1973)

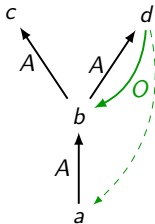
$\mathcal{F} \cup \{\bar{F}\}$ is an antilogism if and only if $\mathcal{F} \vdash F$, and $\mathcal{F}$ is a minimal set of premises required to prove the formula F .

Example of a syllogism (premise selection task)

Given a consistent knowledge base $\mathcal{KB}$ along with a hypothesis H , select the minimal set of premises to derive H from $\mathcal{KB}$

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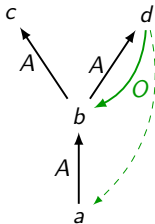


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Type (1) $\{Aab, Odb\} \vdash Oda$

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Experiments (I): Neural models for premise selection

- ▶ **Task:** Given a consistent knowledge base $\mathcal{KB}$ along with a hypothesis H . Train models that provide the necessary premises to derive H , whenever an inference exists.

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- ▶ **Experimental setup:**
 - A single knowledge base.
 - One-hot vector representations of syllogistic formulas.
 - Neural models trained from scratch, including MLPs, RNNs, CNNs, and encoder-only Transformers.

Overall accuracy

Model	Inf.	Best	Mean	SD	NNM
MLP	Val.	93.9	83.2	13.1	88.9
	Inv.	97.1	94.2	2.5	—
	All	96.6	93.5	3.1	—
RNN	Val.	95.9	93.5	1.3	95.3
	Inv.	98.3	97.7	0.5	—
	All	98.0	97.4	0.4	—
CNN	Val.	94.3	92.0	1.3	94.4
	Inv.	97.3	96.7	0.3	—
	All	96.9	96.4	0.2	—
TRA	Val.	96.6	93.6	2.9	95.7
	Inv.	97.8	96.3	1.3	—
	All	97.7	96.1	1.3	—

Overall accuracy by types (MLP)

Inf.	Best	Mean	SD	NNM	HD
1	80.6	46.8	17.9	55.5	1.2
2	92.0	80.9	13.2	83.3	1.1
3	99.2	89.6	8.1	92.9	1.0
4	93.0	78.9	15.9	88.7	1.1
5	100.0	95.0	8.3	96.8	1.0
6	100.0	89.5	11.2	90.0	1.0
7	99.4	88.1	15.4	93.4	1.0
Val.	93.9	83.2	13.1	88.9	1.1
Inv.	97.1	94.2	2.5	—	—
All	96.6	93.5	3.1	—	—

Overall accuracy by types (RNN)

Inf.	Best	Mean	SD	NNM	HD
1	77.8	58.6	9.7	60.7	1.0
2	96.5	92.1	3.6	92.8	1.2
3	99.3	96.6	1.8	96.9	1.0
4	96.1	92.0	2.2	96.1	1.1
5	100.0	99.5	1.2	99.5	0.0
6	100.0	97.7	2.0	97.7	0.0
7	99.4	97.9	1.1	98.9	1.0
Val.	95.9	93.5	1.3	95.3	1.1
Inv.	98.3	97.7	0.5	—	—
All	98.0	97.4	0.4	—	—

Overall accuracy by types (CNN)

Inf.	Best	Mean	SD	NNM	HD
1	78.4	58.3	11.3	64.3	1.7
2	96.6	92.5	2.5	93.1	1.2
3	96.4	92.3	3.4	92.5	1.0
4	93.0	90.6	1.4	96.2	1.1
5	100.0	97.9	2.9	98.4	1.0
6	100.0	93.3	5.8	93.3	0.0
7	100.0	98.3	1.3	99.0	1.0
Val.	94.3	92.0	1.3	94.4	1.2
Inv.	97.3	96.7	0.3	—	—
All	96.9	96.4	0.2	—	—

Overall accuracy by types (TRA)

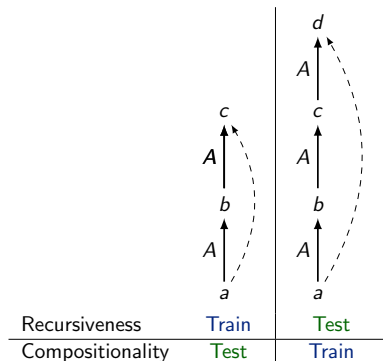
Inf.	Best	Mean	SD	NNM	HD
1	74.1	62.8	8.4	72.1	1.4
2	96.5	90.8	3.6	92.9	1.0
3	96.4	92.6	2.8	94.8	1.3
4	99.6	96.1	3.7	97.9	1.0
5	100.0	94.3	5.2	97.0	1.0
6	100.0	98.6	2.3	99.3	1.0
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Generalization tests for neural models

- ▶ Good generalization (the ability to perform on new data) is an essential aspect of NLP neural models.

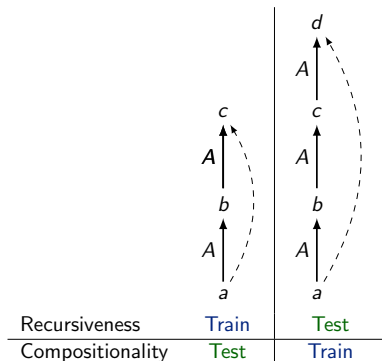
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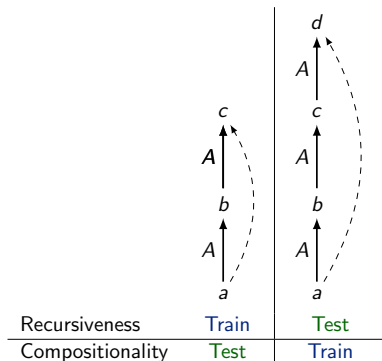
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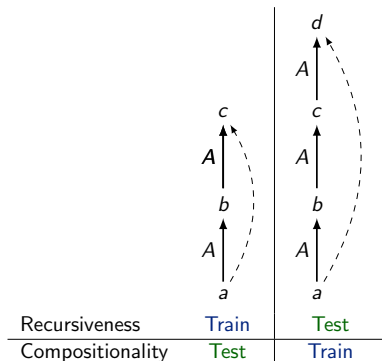
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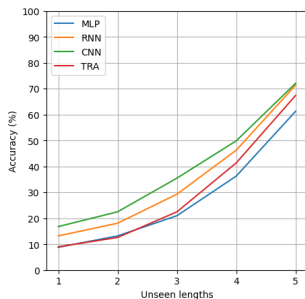
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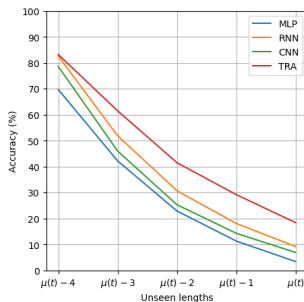
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- ▶ For **test data**, we evaluate the eliminated inferences.

Results

Compositionality



Recursiveness



- **Neural models generalization:** The models cannot learn the logic's fully recursive and compositional nature.

Results

- ▶ **Additional compositional tests:** When certain components of the knowledge base (e.g., segments of A-chains or entire syllogism types) are removed during training, models fail to recognize or generalize to them at test time.

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- ▶ **Additional compositional tests:** When certain components of the knowledge base (e.g., segments of A-chains or entire syllogism types) are removed during training, models fail to recognize or generalize to them at test time.
- ▶ **Basic properties:** The models generalize basic non-compositional and non-recursive features of the syllogistic logic: Principle of Contradiction (either H or $\overline{H}$ is invalid), non-empty denotations of constants (if Aab is valid, then Iab is valid), as well as the symmetry of formulas Iab and Eab .

Experiments (II): Neural models for premise selection and proof by contradiction

- **Task:** Given a consistent knowledge base $\mathcal{KB}$ along with a hypothesis H . Train models that (1) provide the necessary premises to derive H and (2) generate formulas that yield a contradiction, enabling indirect (reductio ad absurdum) proofs.

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- ▶ **Experimental setup:**
 - Multiple knowledge bases.
 - Textual representations of syllogistic formulas.
 - Fine-tuning pre-trained language models, including a relatively small encoder-decoder model (T5) and a substantially larger decoder-only model (GPT).

Accuracy scores for premise selection and proof by contradiction tasks

Experiment	Premise Selection		Proof By Contradiction	
	T5	GPT	T5	GPT
OVE-ALL	0.94 ± 0.05	0.94 ± 0.05	0.93 ± 0.04	0.95 ± 0.04
OVE-SHORT	0.99 ± 0.01	0.98 ± 0.01	0.99 ± 0.01	0.98 ± 0.02
OVE-LONG	0.79 ± 0.17	0.83 ± 0.15	0.76 ± 0.15	0.88 ± 0.15
COM	0.84 ± 0.04	0.76 ± 0.04	0.67 ± 0.05	0.85 ± 0.05
REC	0.80 ± 0.15	0.82 ± 0.15	0.71 ± 0.18	0.86 ± 0.17

T5 Accuracy scores by types (premise selection)

τ	OVE-all	OVE-short	OVE-long	COM	REC
1	0.90 $\pm$ 0.10	0.96 $\pm$ 0.05	0.58 $\pm$ 0.38	0.35 $\pm$ 0.11	0.59 $\pm$ 0.35
2	1.00 $\pm$ 0.01	1.00 $\pm$ 0.00	0.97 $\pm$ 0.09	0.99 $\pm$ 0.01	0.96 $\pm$ 0.08
3	0.85 $\pm$ 0.13	0.96 $\pm$ 0.05	0.61 $\pm$ 0.37	0.88 $\pm$ 0.06	0.66 $\pm$ 0.33
4	0.97 $\pm$ 0.03	1.00 $\pm$ 0.01	0.77 $\pm$ 0.21	0.87 $\pm$ 0.06	0.74 $\pm$ 0.23
5	0.72 $\pm$ 0.28	0.71 $\pm$ 0.36	0.65 $\pm$ 0.41	0.49 $\pm$ 0.37	0.69 $\pm$ 0.38
6	0.97 $\pm$ 0.08	0.99 $\pm$ 0.03	0.84 $\pm$ 0.33	0.90 $\pm$ 0.05	0.88 $\pm$ 0.28
7	0.96 $\pm$ 0.08	1.00 $\pm$ 0.01	0.76 $\pm$ 0.31	0.77 $\pm$ 0.12	0.80 $\pm$ 0.28

GPT Accuracy scores by types (premise selection)

τ	OVE-all	OVE-short	OVE-long	COM	REC
1	0.84 $\pm$ 0.15	0.91 $\pm$ 0.10	0.54 $\pm$ 0.36	0.20 $\pm$ 0.11	0.52 $\pm$ 0.38
2	1.00 $\pm$ 0.01	1.00 $\pm$ 0.00	0.98 $\pm$ 0.06	0.89 $\pm$ 0.04	0.97 $\pm$ 0.06
3	0.87 $\pm$ 0.13	0.98 $\pm$ 0.04	0.67 $\pm$ 0.36	0.87 $\pm$ 0.07	0.68 $\pm$ 0.35
4	0.97 $\pm$ 0.04	0.99 $\pm$ 0.02	0.81 $\pm$ 0.22	0.76 $\pm$ 0.05	0.78 $\pm$ 0.24
5	0.84 $\pm$ 0.23	0.93 $\pm$ 0.17	0.70 $\pm$ 0.38	0.93 $\pm$ 0.16	0.68 $\pm$ 0.39
6	0.99 $\pm$ 0.04	1.00 $\pm$ 0.02	0.94 $\pm$ 0.19	0.92 $\pm$ 0.02	0.92 $\pm$ 0.23
7	0.96 $\pm$ 0.09	0.99 $\pm$ 0.03	0.86 $\pm$ 0.25	0.71 $\pm$ 0.12	0.86 $\pm$ 0.24

T5 accuracy scores by types (proof by contradiction)

τ	OVE-all	OVE-short	OVE-long	COM	REC
1	0.97 $\pm$ 0.05	0.98 $\pm$ 0.04	0.88 $\pm$ 0.21	0.23 $\pm$ 0.12	0.82 $\pm$ 0.27
3	0.88 $\pm$ 0.09	0.98 $\pm$ 0.04	0.54 $\pm$ 0.33	0.95 $\pm$ 0.05	0.37 $\pm$ 0.33
4	0.97 $\pm$ 0.03	1.00 $\pm$ 0.00	0.78 $\pm$ 0.20	0.66 $\pm$ 0.06	0.77 $\pm$ 0.22
5	0.78 $\pm$ 0.29	0.68 $\pm$ 0.39	0.78 $\pm$ 0.33	0.36 $\pm$ 0.36	0.84 $\pm$ 0.31
7	0.95 $\pm$ 0.07	0.99 $\pm$ 0.03	0.85 $\pm$ 0.25	0.83 $\pm$ 0.12	0.80 $\pm$ 0.29

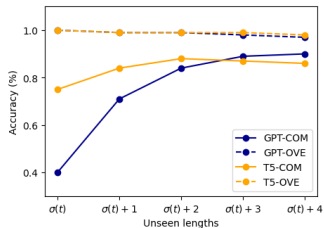
GPT accuracy scores by types (proof by contradiction)

τ	OVE-all	OVE-short	OVE-long	COM	REC
1	0.97 $\pm$ 0.06	0.96 $\pm$ 0.07	0.93 $\pm$ 0.20	0.90 $\pm$ 0.14	0.90 $\pm$ 0.24
3	0.92 $\pm$ 0.12	0.97 $\pm$ 0.08	0.81 $\pm$ 0.30	0.97 $\pm$ 0.05	0.74 $\pm$ 0.34
4	0.97 $\pm$ 0.04	0.99 $\pm$ 0.02	0.87 $\pm$ 0.19	0.78 $\pm$ 0.05	0.81 $\pm$ 0.23
5	0.85 $\pm$ 0.30	0.78 $\pm$ 0.37	0.96 $\pm$ 0.16	0.73 $\pm$ 0.37	0.93 $\pm$ 0.22
7	0.97 $\pm$ 0.07	0.98 $\pm$ 0.06	0.92 $\pm$ 0.19	0.98 $\pm$ 0.04	0.95 $\pm$ 0.15

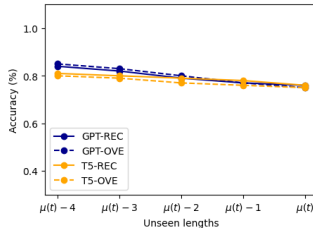
Generalization performance of GPT and T5

Task: Premise selection

Compositionality

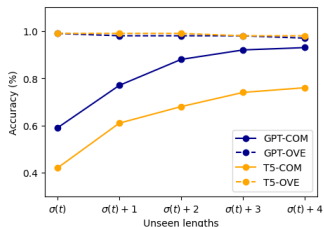


Recursiveness

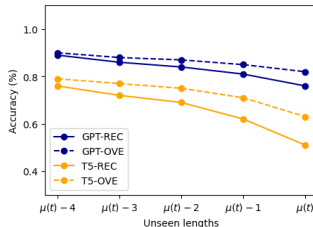


Task: Proof By Contradiction

Compositionality



Recursiveness



Error analysis: unnecessary premises

Accuracy gain from non-minimal proofs and number of unnecessary premises for shorter unseen lengths (COM).

Length	T5			GPT		
	Gain	# Unnec.	Prem.	Gain	# Unnec.	Prem.
$\sigma(\tau)$	0.08	5.55 $\pm$ 4.66		0.26	5.51 $\pm$ 3.53	
$\sigma(\tau) + 1$	0.02	5.77 $\pm$ 4.11		0.10	5.33 $\pm$ 3.86	
$\sigma(\tau) + 2$	0.00	4.76 $\pm$ 4.10		0.04	5.28 $\pm$ 3.63	
$\sigma(\tau) + 3$	0.01	4.40 $\pm$ 3.79		0.02	5.31 $\pm$ 3.78	
$\sigma(\tau) + 4$	0.00	5.22 $\pm$ 4.87		0.02	5.30 $\pm$ 3.94	
Total	0.02	5.44 $\pm$ 4.48		0.09	5.42 $\pm$ 3.66	

Error analysis: unnecessary premises

Accuracy gain from non-minimal proofs and number of unnecessary premises for longer unseen lengths (REC).

Length	T5			GPT		
	Gain	# Unnec.	Prem.	Gain	# Unnec.	Prem.
$\mu(\tau) - 4$	0.00	5.21 $\pm$ 3.48		0.02	4.22 $\pm$ 4.22	
$\mu(\tau) - 3$	0.01	6.86 $\pm$ 3.58		0.02	4.21 $\pm$ 4.16	
$\mu(\tau) - 2$	0.00	7.46 $\pm$ 3.77		0.02	3.48 $\pm$ 3.47	
$\mu(\tau) - 1$	0.00	2.75 $\pm$ 1.30		0.02	3.46 $\pm$ 3.69	
$\mu(\tau)$	0.00	5.67 $\pm$ 1.25		0.01	3.86 $\pm$ 4.69	
Total	0.00	6.06 $\pm$ 3.63		0.02	3.98 $\pm$ 4.04	

Error analysis: missing premises

- ▶ **Term overlap:** checks if hypothesis terms appear in predicted premises (basic syllogistic understanding).

Error analysis: missing premises

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Error analysis: missing premises

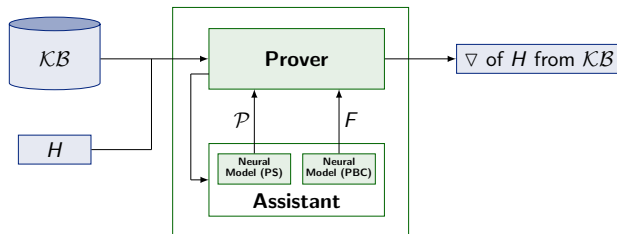
- ▶ **Term overlap:** checks if hypothesis terms appear in predicted premises (basic syllogistic understanding).
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- ▶ **Term validity:** ensures premise terms are limited to those in the knowledge base.

Error analysis: missing premises

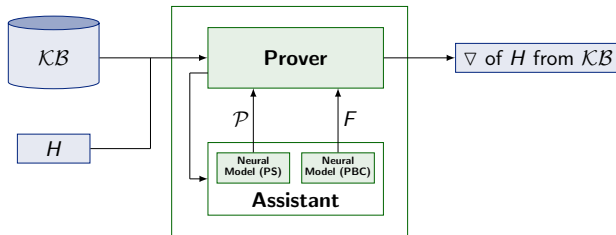
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- ▶ **Premise validity:** verifies predicted premises come from the knowledge base.
- ▶ **Term validity:** ensures premise terms are limited to those in the knowledge base.

Model	Exp.	Term overlap	Prem. valid.	Term val.
T5	OVE	0.77	0.35	0.92
	COM	0.46	0.25	0.90
	REC	0.63	0.18	0.89
GPT	OVE	0.94	0.20	0.92
	COM	0.71	0.31	0.94
	REC	0.92	0.32	0.92

Components of a hybrid model

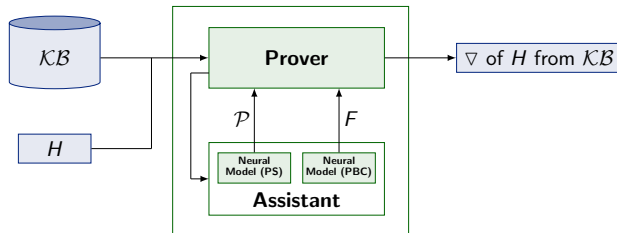


Components of a hybrid model



Input: A knowledge base $\mathcal{KB}$ (set of premises) and a hypothesis H .

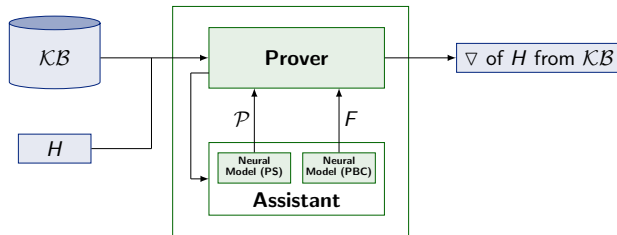
Components of a hybrid model



Input: A knowledge base $\mathcal{KB}$ (set of premises) and a hypothesis H .

Hybrid Model: If the prover asks for assistance, the neural model (PS) provides $\mathcal{P} \subset \mathcal{KB}$ s.t. $\mathcal{P} \vdash H$; and the neural model (PBC) predicts a formula F s.t. $\mathcal{KB} \cup \{\bar{H}\} \vdash F$ and $\mathcal{KB} \vdash \bar{F}$.

Components of a hybrid model

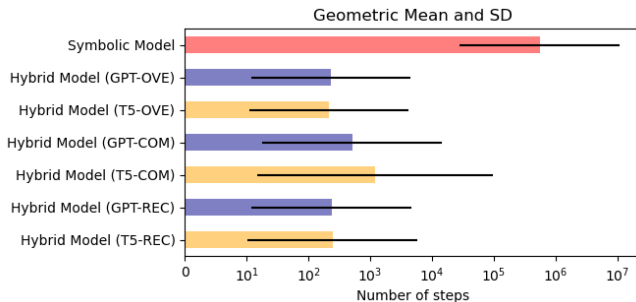


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Output: The prover computes a derivation ∇ (if exists) of H from $\mathcal{KB}$.

Number of steps for the Symbolic and Hybrid models



Conclusions

- ▶ **Neural models generalization:** Pre-trained language models handle recursive reasoning but show weak compositional generalization: training on complex inferences does not transfer to recognizing their simpler components.

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Conclusions

- ▶ **Neural models generalization:** Pre-trained language models handle recursive reasoning but show weak compositional generalization: training on complex inferences does not transfer to recognizing their simpler components.
- ▶ **Hybrid models comparison:** Hybrid models reduce proof steps by approximately three orders of magnitude compared to a purely symbolic model.
- ▶ **Robustness:** Despite limitations in generalization and scale, LLMs remain effective assistants to symbolic provers.

Future work

- ▶ **Extend the logic:** Future work will investigate richer logical fragments, including those studied by Pratt-Hartmann (2004) and selected fragments of modal logic.

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- ▶ **Extend the logic:** Future work will investigate richer logical fragments, including those studied by Pratt-Hartmann (2004) and selected fragments of modal logic.
- ▶ **Generalization analysis:** Studying richer logical systems may reveal new and qualitatively different generalization challenges.

References I

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